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DEPARTMENT OF ELECTRONICS AND TELE-COMMUNICATION ENGINEERING

End Semester Examination May - 2016

Subject : Microprocessor Semester : IV Max. Marks : 70

Date : 7/5/2016 Time : 10 to 1

INSTRUCTIONS TO THE STUDENTS:-

- ***Assume suitable data wherever is necessary.***
- ***Question 1 is mandatory. Attempt any 5 questions from remaining questions.***
- ***Figures to the right in square bracket indicates full marks.***

Q.1 Choose correct answer from the given alternatives **[10]**

I. In an 8085 microprocessor, the instruction CMP B has been executed while the content of the accumulator is less than that of register B. As a result,
   a. Carry flag will be set but Zero flag will be reset
   b. Carry flag will be reset but Zero flag will be set
   c. Both Carry flag and Zero flag will be reset
   d. Both Carry flag and Zero flag will be set

II. An 8085 microprocessor executes "STA 1234H" with starting address location 1FFEH. While the instruction is fetched and executed, the sequence of values written at the address pins A8 - A15 is,
   a. 1FH, 1FH, 20H, 12H
   b. 1FH, FEH, 1FH, FFH, 12H
   c. 1FH, 1FH, 12H, 12H
   d. 1FH, 1FH, 12H, 20H, 12H

III. How many bits are required to store the STAX D instruction?
   a. 2 b. 8 c. 16 d. 1

IV. After acknowledgement of RST5.5 interrupt microprocessor jumps to location,
   a. 002CH b. 0028H c. 0014H d. None

V. After execution of following program what will be the content of stack pointer?

```
LXI SP, 1008H
LXI H, 2008H
SPHL
PUSH PSW
HLT
```

   a. 2006H b. 1006H c. 200AH d. 100AH

VI. While executing STA 2000H instruction value written on $\overline{RD}$ pin of microprocessor during 8th T state is,

a. 1 b. 0 c. 0 – 1 d. 1 – 0

VII. Addressing mode of the SPHL instruction is,

a. Register b. Indirect c. Direct d. Immediate

VIII. How many machine cycles are required for the execution of PCHL instruction?

a. 1 b. 3 c. 2 d. 4

IX. If only RST 7.5 is masked, what will be the status of D1 & D0 bits of an accumulator after execution of RIM instruction

a. 10 b. 00 c. 01 d. 11

X. Byte read from external memory during first machine cycle of every instruction goes to

a. Accumulator b. Temp. register c. Address Buffer d. Instruction register

Q. 2 Attempt any two questions.

I. Assume that 10 numbers are stored at memory locations from 1000H to 1009H. Write an assembly language program to add the numbers at even memory locations only. Store higher byte of a 16 bit result at memory location 2000H and lower byte at memory location 2001H. **[6]**

II. Assume that 10 numbers are stored at memory locations from 1000H to 1009H. Write an assembly language program to find out how many numbers have their D7 bit equal to 1. Store count at memory location 2000H. **[6]**

III. Explain any three addressing modes with suitable examples of each. **[6]**

Q. 3 Attempt any two questions.

I. Write an assembly language program to multiply the lower and upper nibble of a number stored at memory location 1000H. Create a subroutine for multiplication process. **[6]**

II. Find out the time required for the execution of **STAX D** instruction for the oscillator frequencies (a) 6MHz & (b) 4 MHz. **[6]**

III. Draw pin diagram of 8085 microprocessor and explain functions of any three pins. **[6]**

Q. 4 Attempt any two questions

I. Explain following instructions with suitable examples of each

(a) STAX B (b) RET (c) CPI 09H **[6]**

II. Assume that pulse generator is interfaced with RST 7.5 pin of 8085 microprocessor. Write an assembly language program to design a digital clock such that register H will indicate minutes and register L will indicate seconds. Assume that output frequency of pulse generator is 1 Hz. Mask RST6.5 and RST5.5 interrupts. **[6]**

III. Write an assembly language program to receive 8 bit data serially through SID pin of a microprocessor. Find out 2's compliment of that number and store result at 2000H memory location using PUSH instruction. **[6]**

Q.5 What will be the content of Accumulator, Stack pointer and HL pair registers after execution of following program. Also find out the addresses associated with all 3 labels. Assume that program is stored from memory location 0000H (i.e. Starting address of program is 0000H) [12]

```
                LXI H,0FFFH
                LXI SP,2002H
                MVI E, 01H
                MVI D, 00H
                DAD D
                PUSH PSW
                SPHL
                MVI C, 08H
lelelelele :    RIM
                DCR C
                JNZ lelelelele
                LXI H,002BH
tinkatinka:     STC
                RAL
                MOV B, A
                INR C
                MOV A, C
                CPI 09H
                MOV A, B
                JNZ tinkatinka
                PCHL
                MOV H, A
                PUSH H
                MOV A, M
dntwasteurtime : CMA
                DCR L
                JNZ  dntwasteurtime
                HLT
```

Q.6 **Draw timing diagrams for the following instructions.** [12]

I. ADD M

II. MVI A,09H

III. CMA

Q.7 Interface 8 Kbyte of EPROM and 8 Kbyte of RAM with 8085 microprocessor. [12]

*******************************ALL THE BEST*********************************